

Tutorial 4: Vision and Language Models for Situated Language Understanding

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Instructions

- Attempt all exercises before the tutorial session.
- These exercises are designed to encourage critical thinking and application of the concepts covered in the lectures. Focus on the reasoning behind your answers.
- Collaboration is encouraged, but think of your own solutions first.

Exercise 1. Vision-Language Models (VLMs) fundamentally consist of a pre-trained Vision Encoder, a pre-trained Large Language Model (LLM), and a third component, often referred to as a projector or connector, that aligns the visual features with the language model's token space. Discuss the necessity and technical challenges of this *connector* layer. Some points to consider in your discussion:

- What are the possible architectural choices for the connector layer?
- How do you think the choice of connector will affect the stability of training process?
- What are the implications of the connector design on the model's performance (e.g., accuracy/computational cost) in downstream tasks?

Exercise 2. In this exercise, we will design a Vision-Language Model (VLM) system for a real-world application: assistive technology for visually impaired users. The goal is to help users understand their surroundings by describing scenes, reading text in the environment, and answering questions about what is visible through a wearable camera.

- Outline the main components and data flow of your system, specifying how the vision and language modules interact.
- Think about the tasks that your VLM will need to perform and the type of data required for training
- Discuss the challenges you would face in deploying such a system in the real world (e.g., latency, robustness, privacy, user interface).
- Discuss with your peers any implications related to the challenges you identified.

It is reasonable and expected that you will make some simplifying assumptions when solving this exercise. Make sure to state them clearly.